P-P-101E
October 3, 1973
SUPERSEDING
Fed. Spec. P-P-101D
April 6, 1967

FEDERAL SPECIFICATION

PAPER, ABRASIVE, SILICON CARBIDE, WATERPROOF

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers abrasive paper, waterproof, coated on one surface with silicon carbide abrasive grain.

1.2 Classification. Waterproof abrasive paper covered by this specification shall be of one type and in the following grit sizes, as specified (see 6.2):

Grit sizes: 600, 500, 400, 360, 320, 280, 240, 220, 180, 150, 120, 100, 80, and 60.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specification:

UU-P-31 - Paper, General Specifications and Methods of Testing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-3816 - Abrasives and Abrasive Products for Shipment and Storage, Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

FSC 5350

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specified (see 6.2) *

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Commerce Product Standard:

PS8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U S Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive.

3.1.1.1 Silicon carbide. The abrasive grain shall be crushed silicon carbide, of high quality, and free from extraneous materials.

3.1.1.2 Grading, control, and tolerances. Grading of abrasive grain shall be in accordance with PS8 and the tolerances therein shall apply to the abrasive grain recovered from the coated abrasive product.

3.1.2 Backing. The backing shall be made of specialized technical paper manufactured specifically for the purpose. The paper shall be treated with a saturant during manufacture which will render the paper waterproof. In addition, the manufacturer may apply additional coatings to either or both sides of the paper to provide desirable characteristics for abrasive coating, or for end use applications. These additional coatings shall not be considered as part of the backing weight. The backing weight shall meet the requirements of Table I directly after being conditioned at 70°F ± 2°F and fifty plus or minus two percent relative humidity for at least 24 hours. The backing shall be pliable, tough and durable, and shall resist tearing when used under ordinary service conditions.

3.1.2.1 Tensile breaking strength. The average dry breaking strength of the coated abrasive paper shall be in accordance with Table I when tested as described in 4.4.2.

3.1.3 Adhesive. The adhesive shall be of a good quality drying oil, varnish, or other suitable adhesive which will minimize dislodging of the grain from the backing under hard usage when the paper is used wet or dry.

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TABLE I.

GRIT SIZES, BACKING WEIGHT, AND TENSILE STRENGTH OF THE COATED ABRASIVE PAPER

	Nominal Backing Weight 480 Sheets 24x36 Inches		Minimum Average Breaking Strength	
	Weight[1/] + or - 15 Percent	Weight Designation	Machine Direction	Cross Machine Direction
600,500,400,360,320,280 240,220	Pounds 48	A	Pounds 20	Pounds 12
400,360,320,280,240,220 180,150,120,100,80,60	80	C	40	20

1/The basis weight of the paper is calculated in pounds per ream of 480 sheets (includes paper and saturant) 24 by 36 inches at 70°F ± 2°F and 50 ± 2 percent relative humidity.

3.1.4 Coating. The waterproof paper abrasive grain coating shall be "closed coat". The abrasive grain shall be applied to one side of the paper backing only and shall completely cover the surface of the backing (see 6.4).

3.2 Form and dimensions. Sheet - unless otherwise specified (see 6.2) waterproof abrasive paper shall be furnished in 9 by 11 inch sheets, with a tolerance of plus or minus 1/8 inch in length and plus or minus 1/16 inch in width, when measured at and after conditioning at 70°F plus or minus 2°F and 50 percent, plus or minus 2 percent relative humidity for at least 24 hours.

3.3 Identification marking. Each sheet of abrasive paper shall be marked in a legible and permanent manner with the following data:

a. Manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

b. The name of the abrasive or the manufacturer's trade designation of such known character for the abrasive that the abrasive grain may be readily determined.

c. The grit size representing the grain size used.

d. The word "waterproof" or a trademark which will readily distinguish the product as being waterproof.

e. The backing weight designation.

3.4 Workmanship. The workmanship shall be of the highest grade and equal or better in every respect to that which is considered good practice by industry.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that suppliers and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. All abrasive paper of the same backing weight and grit size manufactured under essentially the same conditions and offered for delivery at one time shall be considered a lot for the purposes of examination and tests.

4.2.2 Sampling for examination of packaging and packing. Packaged and packed abrasive paper in shipping containers shall be sampled in accordance with Inspection Level I for conformance to Section 5.

4.2.3 Sampling for visual, dimensional, and workmanship characteristics of end items.

a. A random sample of filled packages shall be selected from each lot of filled packages offered for delivery in accordance with inspection Level I.

b. Five sheets shall be selected at random from each sample package for visual and dimensional examination. Each sheet shall be selected from a different folder.

4.2.4 Sampling for tests. Samples for test shall be selected in accordance with inspection Level I. Samples may be selected from those selected in accordance with 4.2.3. One sheet shall be selected for each of the tests in 4.4 except when necessary to select additional sheets. Each sheet shall be selected from a separate package.

4.3 Examination. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 Examinations.

4.3.1.1 Packaged and packed abrasive paper. Abrasive paper ready for shipment, selected in accordance with 4.2.2 shall be examined with AQL's and classification of defects in accordance with MIL-P-3816.

4.3.1.2 Individual sheets. Samples of abrasive paper selected in accordance with 4.2.3 shall be examined to verify conformance to all requirements of this specification, including workmanship, which does not involve tests. The sheets shall be individually examined for conformance to size after conditioning (see 3.2), marking, grit coverage, and various elements of workmanship which includes straight, clean cut edges and square corners. The surface on which the grit is coated shall be carefully examined for such defects as foreign matter, uneveness of or voids in the coating, loose grit, backing cracks, or other defects which may impair (major defects) the usefulness of the abrasive paper. The AQL's for defects described above shall be 2.5 percent defective.

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4.4 Tests. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.4.1 Testing for grit size.

4.4.1.1 Grading for grit size. Grading to determine the grit size, including recovery of abrasive grain from the coated abrasive product, preparation for grading, and procedures shall be in accordance with Product Standard PS8.

4.4.1.2 Certification of grading. Grading tests specified in 4.4.1.1 may be omitted in instances where the contractor produces certification in writing that the abrasive grain complies with the requirements of PS8 for silicon carbide. Contractors are urged to include this certification with material offered for inspection and inspectors are urged to determine whether such certification is included with such material in order to preclude the necessity for grading tests.

4.4.2 Breaking strength testing. The tensile breaking strength of the coated abrasive paper directly after conditioning for at least 24 hours at 70°F plus or minus 2°F and 50 percent, plus or minus 2 percent relative humidity, shall be determined by Method 171 of UU-P-31, except that the jaws of the testing machine shall be five inches apart, and that six specimens one inch wide and seven inches long shall be taken both longitudinally (machine direction) and transversely (cross machine direction) from sample sheets. The average tensile strength in each direction shall be computed from at least three specimens. Samples tearing in the jaws of the tensile testing machine should be discarded (see 3.1.2 and Table I).

4.4.3 Practical service testing. At least one sheet of waterproof coated abrasive paper shall be subjected to a practical service test of short duration to determine whether the paper is suitable for the service intended. Water shall be used as a lubricant in this test.

5. PREPARATION OF DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing of abrasive paper shall be in accordance with MIL-P-3816, as specified for levels A, B, and C (see 6.2). The number of sheets per folder shall be as specified (see 6.2). Unless otherwise specified, the number of sheets per folder will be 50.

5.2 Palletization. Unless otherwise specified (see 6.2) all paper preserved, packaged and packed as specified in 5.1 shall be palletized in accordance with MIL-P-3816.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, packages and shipping containers shall be marked in accordance with Federal Standard No 123 (see 6.2).

5.3.2 Military activities. In addition to any special markings required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

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6. NOTES

6.1 Intended use. Waterproof abrasive paper is used principally for "sanding" or finishing wood, metal, plastic, paint, enamel, lacquer, and other similar material. The finer grit sizes (600 to 220) are generally used in finishing operations whereas the coarser grit sizes (180 to 60) are generally used where fast stock removal is desired. Waterproof abrasive paper is normally used with water as a lubricant. However, some finishing operations require oils or special waxes as lubricants and these may be used equally as well. Waterproof abrasive paper may be used by hand or it may be cut into strips and used on light duty sanding machines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Grit size required (see 1.2 and Table I).

c. Paper weight designation required for grit sizes 400 through 220 (see Table I).

d. Selection of applicable levels of packaging and packing required (see 5.1).

e. Number of sheets per folder required (see 5.1).

f. Any special marking required (see 5.3.1 and 5.3.2).

g. Whether civil or military marking is required (see 5.3).

h. When palletization is not required (see 5.2).

i. Size of sheet required if other than 9 by 11 inches (see 3.2).

6.3 Silicon carbide. A blue-black crystalline variety of synthetic mineral, an electric furnace product, which in crushing operations develops sharp edges. The hardness is approximately 9.7 (Moh's scale) or 2480 Knoop scale, with a specific gravity of 3.2.

6.4 A closed coat is one in which the abrasive grain is put on in such quantity as to entirely cover the backing.

Custodian:
Army - MR
Air Force - 84

User activities:
Army - MI, ME
Navy - YD, MC, OS, AS

Preparing activity:
Air Force - 84

Project No 5350-0044

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
P-P-101E, Paper, Abrasive, Silicon Carbide, Waterproof

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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